Dynamic Escape Planner During Natural Disasters

Final Review

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Contents

1. Introduction
2. Problem Statement
3. Objectives
4. Existing System
5. Proposed System
6. System Architecture
7. S/W and H/W Requirements
8. Algorithm / Core Techniques
9. Modules
10. Implementation/Screenshots
11. SDG Addressed
12. Conclusion
13. Future Scope
14. References

Introduction

Natural disasters such as floods and landslides can rapidly make normal road navigation unsafe. Conventional GPS systems mainly optimize routes based on distance or travel time, which may lead people toward flooded or low-lying areas. The geographically closest shelter may not always be the safest option during a disaster.

PlanESC models Kerala's transportation system as a weighted graph using real GIS road data.
The system combines:

- OpenStreetMap road networks
- Hydrological hazard information
- Elevation data
- Road capacity
- Graph algorithms
- It dynamically identifies safer evacuation corridors to high-ground sanctuaries.

Problem Statement

To develop a Dynamic Escape Planner that identifies safe evacuation routes during disasters, predicts road closures, reduces traffic congestion, and optimizes the allocation of shelters and emergency resources using real-time data and advanced graph algorithms.

Objectives

- Model Kerala’s road network as a graph using real roads, intersections, and transit points.
- Generate safe evacuation routes to suitable high-ground sanctuaries.
- Consider flood risk, elevation, travel time, and road capacity.
- Avoid flood-prone and hazardous areas during disasters.
- Identify highway bottlenecks and optimize evacuation flow.
- Apply Dijkstra, Kruskal’s MST, and network-flow algorithms for route and network optimization.
- Provide an interactive GIS interface for quick emergency decision-making.

Existing System

Traditional GPS and navigation systems mainly focus on shortest distance, travel time, and normal traffic conditions. During disasters, however, road conditions and safety levels can change rapidly, making conventional navigation unreliable.

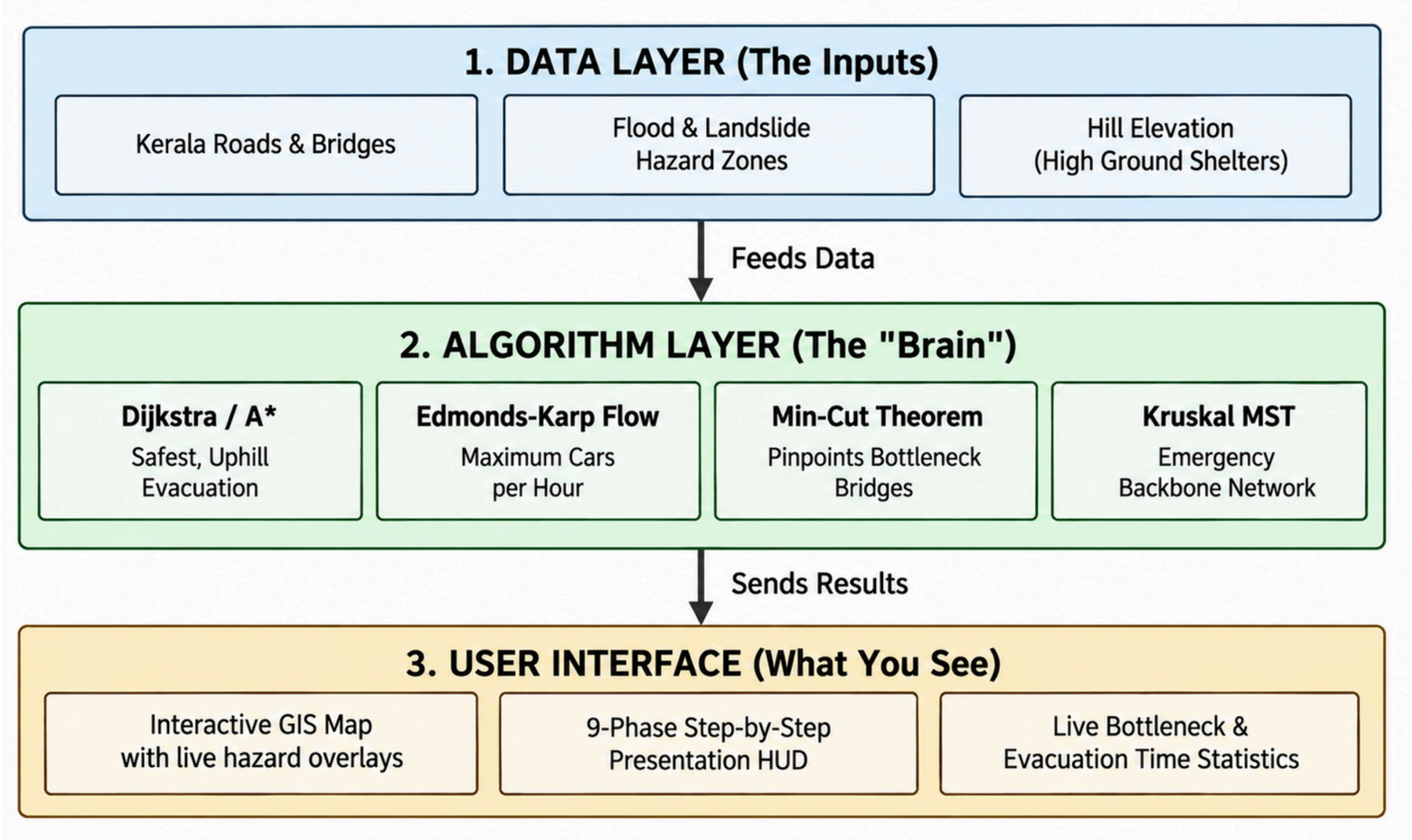
- May suggest flooded or damaged roads
- May select the nearest but unsafe shelter
- Does not consider elevation and flood risk
- Limited road-capacity and bottleneck analysis
- Cannot adapt effectively to rapidly changing disaster conditions

Proposed System

PlanESC is an interactive graph-based disaster evacuation system that uses real Kerala road networks, flood-risk data, elevation, and road capacity to generate safer evacuation routes.

- **GIS Road Network**: Uses OpenStreetMap roads, junctions, and transit points.
- **Hazard-Aware Routing**: Avoids flood-prone and low-lying areas.
- **Safe Sanctuary Selection**: Prioritizes high-ground sanctuaries instead of simply choosing the nearest shelter.
- **Capacity Analysis**: Considers road capacity and identifies potential evacuation bottlenecks.
- **Graph Algorithms**: Uses Dijkstra, Edmonds-Karp, Max-Flow Min-Cut, and Kruskal's MST for route and network optimization.
- **Interactive GIS Interface**: Displays routes, hazards, sanctuaries, and graph visualizations for faster emergency decision-making.

System Architecture



Software Requirements

1. **Software Requirements:**
   - **Backend Environment:** Python 3.8+ (Python 3.13 recommended)
   - **Frontend & GIS:** Html, JavaScript, CSS
   - **Supported OS:** Windows 10/11, Linux (Ubuntu 20.04+), macOS Monterey+.
   - **Map API :** OpenStreetMap
2. **Hardware Requirements:**
   - **Client Workstation:** Dual-core 2.0 GHz CPU, 4 GB RAM.

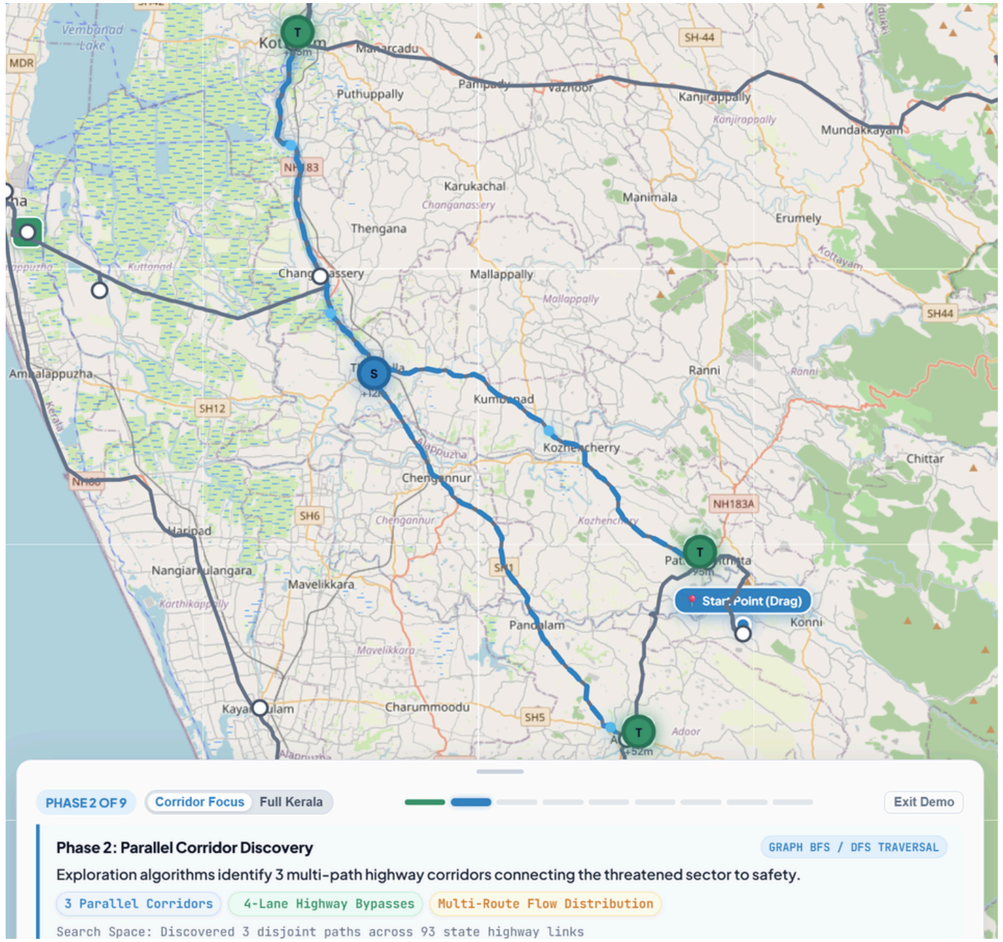
Core Algorithms

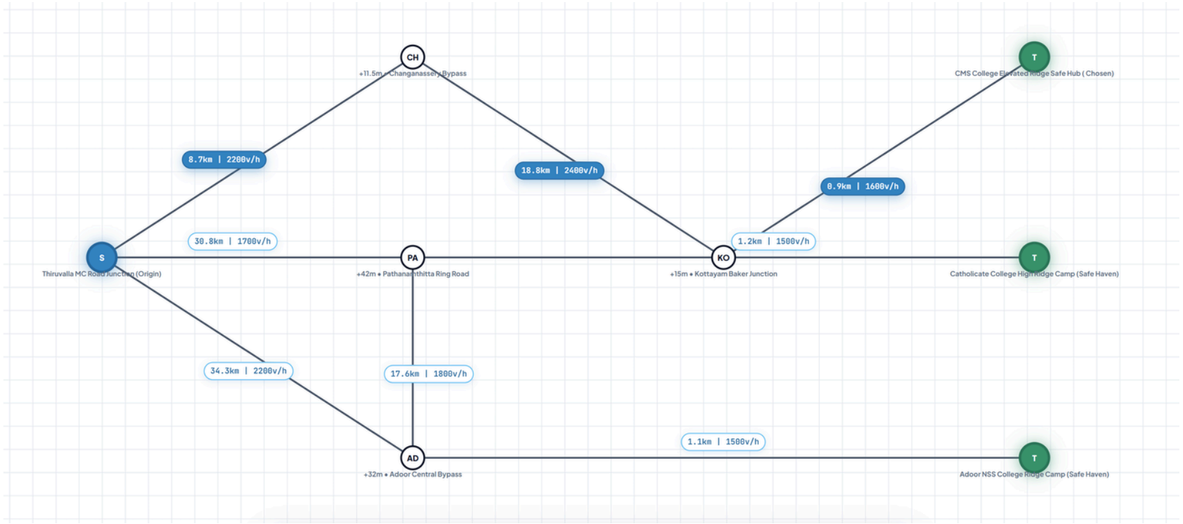
Algorithm	Computational Objective
Dijkstra	Explores network paths in order of cumulative cost g(n) to calculate single-source shortest/safest paths to all accessible nodes.
Edmonds-Karp Max-Flow	Computes maximum total vehicle flow from origin zones to highland sinks using augmenting paths.
Max-Flow Min-Cut Theorem	Identifies the set of bottleneck edges whose removal isolates source from sink with minimum capacity.
MST	Grows a minimum spanning tree sequentially from a designated central command hub node.

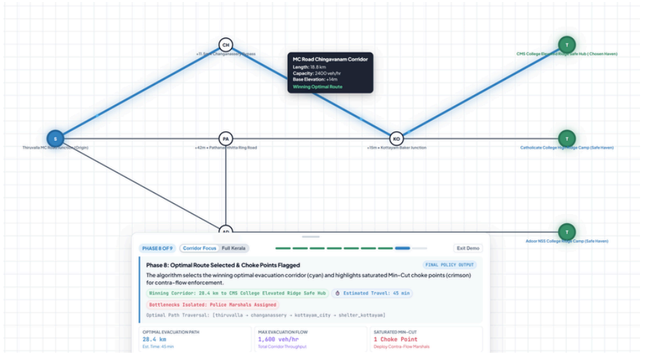
Modules

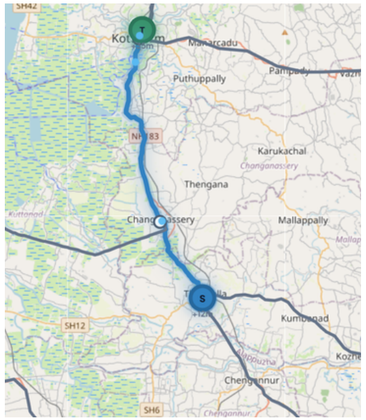
1. GIS Mapping Module:
   - Multi-layer map switcher (Street View, Satellite Reconnaissance, Topographic Contours, Dark EOC Mode).
   - Click-anywhere snapping and dynamic hazard breach simulation.
2. Evacuation Routing Module:
   - Evaluates road elevation, flood proximity, travel time, and vehicular capacities.
   - Computes safest corridor to optimal sanctuary.
3. Network Throughput & Bottleneck Module:
   - Computes maximum vehicle flow rate using Edmonds-Karp.
   - Highlights critical infrastructure links whose failure cuts evacuation paths (Min-Cut).
4. Logistics & Emergency Network Module:
   - Generates Minimum Spanning Trees (MST) using Kruskal/Prim to optimize emergency supply delivery lines.

Implementation/Screenshots









SDG Addressed

SDG 11: Sustainable Cities and Communities

- Target 11.5: "Significantly reduce the number of deaths and the number of people affected and substantially decrease the direct economic losses caused by disasters, with a focus on protecting the poor and people in vulnerable situations."

Work Progress

- Identified the need for graph-theoretic disaster evacuation decision support and safe sanctuary allocation
- Modeled statewide GIS road topologies, elevation profiles, and hydrological flood hazard zones
- Implemented multi-criteria pathfinding, corridor bottleneck capacity analysis, and emergency network spanning algorithms
- Developed a local search engine for multi-stop rescue routing and supply distribution
- Designed an interactive GIS command center featuring dynamic map-to-graph morphing presentation capabilities
- Evaluated routing efficiency, shelter suitability, and flow capacity under simulated disaster scenarios

Future Scope

- Integration of real-time IoT water level sensors and satellite radar telemetry for automated hazard detection
- Machine learning models for dynamic traffic congestion forecasting and proactive contra-flow routing
- Mobile Progressive Web App with offline GPS navigation for citizens during network outages
- Automated SMS emergency alert integration

Conclusion

PlanEsc ridges graph theory and GIS mapping to deliver an intelligent disaster evacuation system. By combining multi-criteria pathfinding, bottleneck flow analysis, and high-ground shelter optimization, the system prevents evacuees from being funneled into flood traps while optimizing traffic and rescue logistics. Operating seamlessly via a Python backend, it offers a practical decision-support tool for disaster management authorities.

References

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